FDE Take-Home Assignment: Technical Statement

While building my RAG Research Extension, I noticed a product gap between /search and /research.

Tavily Research Mini is thorough, but its typical latency of 30–180 seconds makes it difficult to use in conversational products where users expect fast, continuous interaction. /search advanced, by contrast, is fast at roughly 1–5 seconds, but it is too shallow for tasks that require multi-hop synthesis, source reconciliation, and deeply structured outputs.

The core thesis is that customers want structured, citation-backed research with some multi-hop reasoning, but they do not always need the full cost and latency of a complete Research Mini run.

To address this, I built *Tavily Research Mini Lite*: a middle-ground research workflow that is deeper than Search Advanced, but faster and cheaper than Research Mini.

Research Mini Lite uses a bounded LangGraph agent loop that:

1. Decomposes the query into a small set of focused sub-questions.
2. Runs parallelized, single-hop Tavily /search calls.
3. Aggregates and deduplicates the retrieved evidence.
4. Performs post-hoc synthesis and structured extraction.
5. Returns a structured, citation-backed response (or custom output json schema).

This design avoids the open-ended latency of a full research agent while still producing synthesized, multi-source answers. The goal is to hit a target latency of approximately 15 seconds, making the system fast enough for user-facing applications while providing substantially more depth than standard search.

From a technical perspective, *Research Mini Lite* offers predictable latency, parallel retrieval, structured outputs, citation-backed reasoning, and lightweight multi-hop synthesis. From a business perspective, it creates a practical product tier for browser extensions, AI copilots, customer-facing agents, and other conversational integrations where research needs to feel both intelligent and responsive.

In short, *Research Mini Lite* fills the gap between Search Advanced and Research Mini by giving developers a faster, cheaper, and more interactive way to generate structured research outputs without requiring a full deep-research run.